UNIT 4 · TOPIC 3(B)

Firing Order and Firing Intervals

Syllabus topic: Calculations of power, torque, speed; firing arrangement and intervals

Firing interval

$$\text{4-stroke: firing interval} = 720^\circ \div \text{number of cylinders; 2-stroke: } 360^\circ/K$$

Cylinders (4-stroke)	Firing interval
1	720°
2	360°
3	240°
4	180°
6	120°

Even intervals give smooth torque and balanced bearing loads.

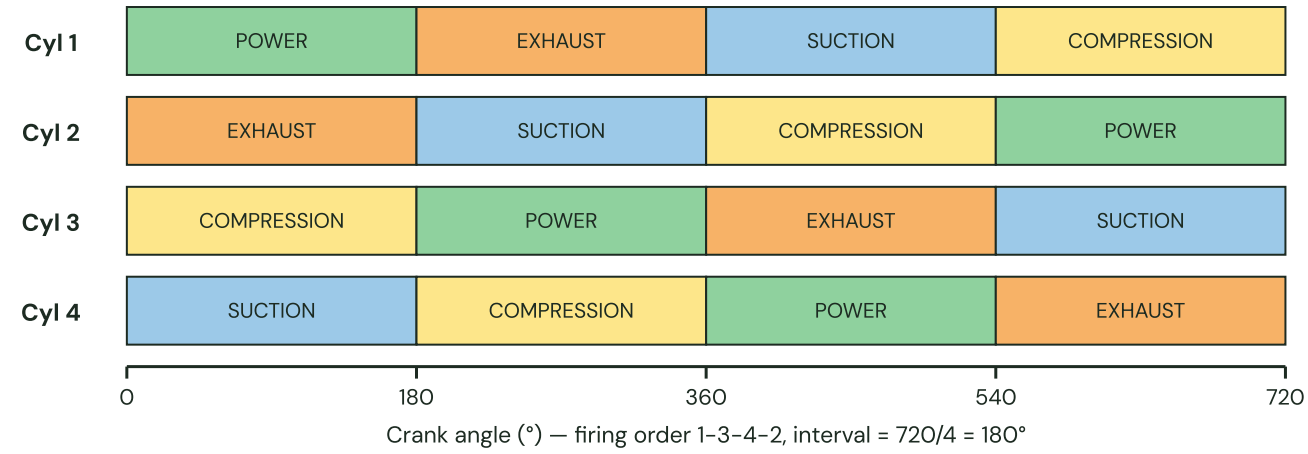
Firing order

Engine	Common firing order
3-cylinder	1-2-3 (or 1-3-2)
4-cylinder	**1-3-4-2** (most Indian tractor engines; alternative 1-2-4-3)
6-cylinder inline	1-5-3-6-2-4

The order is chosen to:

- Distribute successive power impulses along the crankshaft (limit main-bearing load and crank twist).
- Avoid suction clash between adjacent cylinders in the manifold.
- Aid even cooling.

In a 4-cylinder engine the crank throws are paired **1 & 4** and **2 & 3**, 180° apart: pistons 1 & 4 move together, 2 & 3 opposite.



Stroke chart of a 4-cylinder, 4-stroke engine (cranks 1 & 4 up, 2 & 3 down; redrawn with full stroke names)

Reading the stroke chart — the classic exam puzzle

"If cylinder 1 has just fired, which is on exhaust?" Follow the firing order backwards:

1. Cylinder 1 is on POWER (0–180°).
2. Cylinder 3 fires next, so it is on COMPRESSION.
3. Cylinder 2 fired just before cylinder 1 (last in the order), so it is on EXHAUST.
4. Cylinder 4 is on SUCTION.

Practice MCQs

Q81. Firing interval of a 4-stroke, 4-cylinder engine is:

(a) 90° (b) 120°
(c) 180° (d) 240°

Q82. Firing interval of a 4-stroke, 3-cylinder engine is:

(a) 120° (b) 180°
(c) 240° (d) 360°

Q83. Firing interval of a 4-stroke, 6-cylinder engine is:

(a) 60° (b) 90°
(c) 120° (d) 180°

Q84. The most common firing order of 4-cylinder tractor engines is:

(a) 1-2-3-4 (b) 1-3-4-2
(c) 4-3-2-1 (d) 1-4-3-2

Q85. Standard firing order of an inline 6-cylinder engine is:

(a) 1-2-3-4-5-6 (b) 1-5-3-6-2-4
(c) 6-5-4-3-2-1 (d) 1-6-2-5-3-4

Q86. In a 4-cylinder engine, crank throws are paired:

(a) 1&2, 3&4 (b) 1&3, 2&4
(c) 1&4, 2&3 (d) Random

Q87. With order 1-3-4-2, when cylinder 1 is on POWER, cylinder 3 is on:

(a) Exhaust (b) Compression
(c) Suction (d) Power

Q88. And cylinder 2 is on:

(a) Exhaust
(b) Compression
(c) Suction
(d) Power

Q89. Firing orders are chosen mainly to:

(a) Save fuel
(b) Distribute power impulses along the crank and balance bearing/manifold loads
(c) Reduce cost
(d) Ease casting

Q90. In a 2-stroke, 3-cylinder engine, firing interval is:

(a) 240°
(b) 180°
(c) 120°
(d) 360°

Answers

Q81 — (c) 180°
Q82 — (c) 240°
Q83 — (c) 120°
Q84 — (b) 1-3-4-2
Q85 — (b) 1-5-3-6-2-4
Q86 — (c) 1&4, 2&3
Q87 — (b) Compression — *Order 1-3-4-2 at 180° spacing: cylinder 3 fires next, so it is compressing now.*
Q88 — (a) Exhaust — *Cylinder 2 is last in the order — it fired 180° before cylinder 1, so it is now on exhaust (see the stroke chart).*
Q89 — (b) Distribute power impulses along the crank and balance bearing/manifold loads
Q90 — (c) 120°

Sources: Liljedahl et al. — Tractors and their Power Units; standard IC-engine texts; Kepner et al.; NPTEL farm power lectures.